Semantic-Aware Partitioning of Property Graphs

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Abstract. Property graphs are widely used to model richly connected, attributed data. As such datasets outgrow a single machine, they must be partitioned across machines for storage, query processing, and reasoning. Classical partitioners optimise structural objectives such as edge cut and load balance. Property graphs, however, also carry semantic constraints, including keys, functional dependencies, and generating dependencies. These constraints govern identity, consistency, and inference, and a purely structural cut can split them across partitions. This half-day tutorial studies how property graphs can be partitioned without losing such semantics. It is organised in four parts: the property graph model and its constraint languages; the foundations of balanced graph partitioning; constraint-aware partitioning, in which constraints guide placement; and a hands-on demonstration on public datasets that contrasts structural and constraint-aware placement. The tutorial closes with open problems across partitioning, reasoning, and the Semantic Web.

1 Overview and Relevance

Graph partitioning splits a graph into balanced parts while minimising the edges between them, which is a prerequisite for distributing a graph across machines for storage, query processing, and reasoning. Structural partitioning is well studied, from classical multilevel schemes such as METIS [7] to parallel multi-constraint methods [12]. These optimise topological objectives such as edge cut and load balance, treating the graph as a purely structural object. Property graphs, however, can also carry semantic constraints such as key constraints [2], graph functional dependencies [6], or more general graph generating dependencies [13]. Such constraints may span multiple vertices and edges, so graph partitioning can separate the data they depend on. Checking or reasoning with it may require data from more than one partition.

The tutorial explores graph partitioning from a Semantic Web perspective, emphasizing scalability and semantic constraints. The material is organized in four parts. The first introduces property graphs and their constraint languages. The second presents the foundations of graph partitioning. The third covers constraint-aware partitioning, where constraints guide data placement. The fourth provides a hands-on demonstration and discusses open challenges.

The topic is timely for the Semantic Web community. Property graphs are now backed by recent ISO standards: GQL (ISO/IEC 39075:2024) and SQL/PGQ (ISO/IEC 9075-16:2023). Their constraint and schema layer has also matured. PG-Keys [2] and PG-Schema [3] provide constraint and schema languages for property graphs. Recent work also places PG-Schema, SHACL, and ShEx on a common formal foundation [1]. Partitioning property graphs raises the same questions of data placement and constraint preservation that the Semantic Web community already faces with RDF, SPARQL, SHACL, and ShEx, which makes the topic directly relevant to ISWC.

Recent tutorials at database and Semantic Web venues have covered adjacent topics: techniques and systems for querying graph-structured data [11,10], human-centric and visual graph querying [4], knowledge-graph embeddings [15], and property graph standards [8]. To our knowledge, no prior work uses graph constraints to guide property graph partitioning

2 Content and Schedule

The tutorial has four sections of roughly 50 minutes, each with a short discussion or hands-on portion.

Part 1: Property Graphs and Their Constraints. [50 min]
We introduce property graphs, graph querying, and major constraint languages: PG-Keys [2], graph functional dependencies [6], and graph generating dependencies [13], together with their RDF counterparts, SHACL and ShEx [1].
– *Audience interaction.* A short discussion of which integrity constraints real datasets require, and which current PG constraint languages can express.

Part 2: Foundations of Graph Partitioning. [50 min]
We define the balanced graph partitioning problem. We then review the main partitioning models (spectral, multilevel, and streaming), algorithmic approaches, and quality metrics, including balance, communication cost, and replication.
– *Audience interaction.* On a small example, the audience identifies which constraints a structural cut would split across partitions, motivating Part 3.

Part 3: Constraint-Aware Partitioning. [50 min]
This is the core of the tutorial. We present constraint-aware partitioning, where constraints guide data placement alongside graph structure. We compare it with existing partitioning approaches and discuss metrics for measuring constraint fragmentation and cross-partition coordination.
– *Audience interaction.* A discussion of which constraint classes benefit most from co-location and the trade-offs with partition balance.

Part 4: Hands-on Demonstration and Open Challenges. [50 min]
We present a demonstration on public datasets, such as DBLP, CORDIS and GDELT, and benchmarks, such as LDBC SNB [5]. The demonstration contrasts structural and constraint-aware partitioning and discusses open challenges in constraint handling, repartitioning, and reasoning.

– *Audience interaction.* Attendees will run the demonstration on the released materials and reflect on semantic-aware partitioning.

3 Formal Details

Format: The tutorial is intended to be presented in a **half-day format**.

Level: Beginner to intermediate.

Prerequisite knowledge: Basic familiarity with graphs and algorithms is assumed. Knowledge of RDF/SPARQL, SHACL/ShEx, or distributed systems is helpful but not required. No programming is needed to follow the demonstration.

Target audience: Researchers, PhD students, practitioners, and developers working with property graph data and distributed graph processing, with a focus on scalable data placement and constraint preservation in RDF and Semantic Web settings. We estimate 30–60 participants.

Learning outcomes: After the tutorial, attendees will be able to (i) relate property graph constraints to RDF, SHACL, and ShEx; (ii) explain graph partitioning models, algorithms, and metrics; (iii) understand the role of constraint-aware partitioning in preserving semantic context; (iv) assess partitioning with structural and semantic metrics; and (v) identify open challenges in partitioning, reasoning, and the Semantic Web.

Presenter information:

– **Elisjana Ymeralli** is a PhD candidate (https://users.ics.forth.gr/~ymeralli/) at the University of Crete and a Research Fellow at FORTH-ICS, working on big data partitioning and the Semantic Web. She holds a B.Sc. and an M.Sc. in Computer Science from the University of Crete. She has co-authored several publications in international venues, including CAiSE, RuleML+RR, and JURIX and has contributed to tutorial activities at VLDB. She leads Part 3.
– **Georgia Troullinou** is a postdoctoral researcher (https://scholar.google.com/citations?user=DoJS1DsAAAAJ) at ICS-FORTH. Her research interests include knowledge representation, semantic schema discovery, graph summarization, and large-scale data partitioning. She has co-authored numerous publications in leading international venues, including VLDB, EDBT, ISWC, and ESWC, and has contributed to research on RDF data management, and property graph standards. She leads Part 4.
– **Stefania Dumbrava** is an Associate Professor (https://web4.ensiie.fr/~stefania.dumbrava/) at ENSIIE and Télécom SudParis, France, working on reliable graph data processing. She has co-authored over 40 publications, including PG-Schema and PG-Keys, in venues such as SIGMOD, VLDB, and EDBT. She has taught graph databases at VLDB and ACM Summer Schools and co-presented the VLDB 2025 tutorial on Property Graph Standards. She leads Part 1.
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and an Affiliated Researcher at FORTH-ICS. He has authored over 280 publications in leading conferences and journals, including ACM SIGMOD, VLDB, EDBT, ISWC, and ESWC, with more than 50 publications on Knowledge Graphs. He also has extensive experience delivering high-profile tutorials on related topics (VLDB, EDBT, ISWC etc). He leads Part 2.

Materials: All materials will be released before the conference under open licences, including slides, reading materials, datasets, configurations, and reproduction scripts. The demonstration relies on open-source components and benchmark datasets, so attendees can re-run and extend it.

Previous editions: This is the first edition of the tutorial. It builds on the presenters' tutorial experience at database and Semantic Web venues, including VLDB 2025 [8], ISWC 2024 [14] and EDBT 2019 [9], and extends it to the largely unexplored area of property graph partitioning.

References

1. Ahmetaj, S., Boneva, I., Hidders, J., et al.: Common Foundations for SHACL, ShEx, and PG-Schema. In: WWW. pp. 8–21 (2025)
2. Angles, R., Bonifati, A., Dumbrava, S., et al.: PG-Keys: Keys for Property Graphs. In: SIGMOD Conference. pp. 2423–2436 (2021)
3. Angles, R., Bonifati, A., Dumbrava, S., et al.: PG-Schema: Schemas for Property Graphs. Proc. ACM Manag. Data **1**(2), 198:1–198:25 (2023)
4. Bhowmick, S.S., Choi, B., Li, C.: Graph querying meets HCI: state of the art and future directions. In: SIGMOD Conference. pp. 1731–1736 (2017)
5. Erling, O., Averbuch, A., Larriba-Pey, J.L., et al.: The LDBC social network benchmark: Interactive workload. In: SIGMOD Conference. pp. 619–630 (2015)
6. Fan, W., Wu, Y., Xu, J.: Functional dependencies for graphs. In: SIGMOD Conference. pp. 1843–1857. ACM (2016)
7. Karypis, G., Kumar, V.: A fast and high quality multilevel scheme for partitioning irregular graphs. SIAM J. Sci. Comput. **20**(1), 359–392 (1998)
8. Kondylakis, H., Dumbrava, S., Lissandrini, M., et al.: Property Graph Standards: State of the Art & Open Challenges. Proc. VLDB Endow. **18**(12), 5477–5481 (2025)
9. Kondylakis, H., Kotzinos, D., Manolescu, I.: RDF graph summarization: Principles, techniques and applications. In: EDBT. pp. 433–436 (2019), tutorial
10. Manolescu, I., Mohanty, M.: Full-power graph querying: State of the art and challenges. Proc. VLDB Endow. **16**(12), 3886–3889 (2023)
11. Mhedhbi, A., Salihoglu, S.: Modern techniques for querying graph-structured relations: Foundations, system implementations, and open challenges. Proc. VLDB Endow. **15**(12), 3762–3765 (2022)
12. Schloegel, K., Karypis, G., Kumar, V.: Parallel multilevel algorithms for multi-constraint graph partitioning. In: Euro-Par. LNCS, vol. 1900, pp. 296–310 (2000)
13. Shimomura, L.C., Fletcher, G., Yakovets, N.: GGDs: graph generating dependencies. In: CIKM. pp. 2217–2220 (2020)
14. Vrgoˇc, D.: Recent advances in graph data management. https://domagojvrgoc.github.io/ISWC2024-GraphDB/ (2024), (Accessed: November 2024)
15. Xiong, B., Nayyeri, M., Daza, D., Cochez, M.: Reasoning beyond triples: Recent advances in knowledge graph embeddings. In: CIKM. pp. 5228–5231 (2023)